Analytical Cume
February 28, 1998

To routine analytical procedures for determining the concentration of unknowns is that of standard addition and internal standard.

1. (30 points) Internal standards are desirable whenever losses of sample are likely to occur during handing or analysis. This problem is especially apparent in chromatography, where the amount of sample injected is not precisely known, and can vary from injection to injection. An internal standard is a known amount of compound (B), different from the analyte (A), that is added to the analyte. The principle is that signals from the standard and analyte are both linearly related to the concentration of each by proportionally constants or $S_A = k_A C_A$ and $S_B = k_B C_B$, where S is the signal received from concentration C and k is the constant of proportionality. The ratio of these two curves can then be used to determine the concentration on an unknown analyte.

A mixture containing 80.0nM iodoacetone (designated A) and 64.0 nM p-dichlorobenzene (designated B) gave the relative detector response (peak A/peak B) = 0.71 in a chromatography experiment.

A solution containing an unknown quality of A plus 930 nM B gave a relative detector response (peak A/peak B) = 1.21.

Find the concentration of A in the unknown.

2. (70 points) In standard addition, known quantities of the analyte are added to an unknown concentration of the same analyte. The increase in signal lets us deduce the how much analyte was in the original solution. This method also require a linear response of the signal to analyte.

The quantity of acetylene in a gas mixture was measured by mass spectroscopy, in which the signal for mass number 26 is proportional to the volume percent of acetylene.

Gas	Signal (mV)
Blank (no acetylene)	0.2
Unknown	10.8
Unknown + 0.072 volume% C_2H_2	17.1
Unknown + 0.121 volume% C_2H_2	20.2
Unknown + 0.200 volume% C_2H_2	30.0
Unknown + 0.364 volume% C_2H_2	44.6

Derive the equation:

$$\frac{[X]_i}{[S]_f + [X]_f} = \frac{S_x}{S_{x+s}}$$

where X stands for unknown, S for standard (bear in mind that the chemical species X and S are the same) i stands for initial concentration and f for the final diluted concentration of standard or unknown. For example $[X]_f$ stands for the concentration of the unknown the sample after adding the extra volume of the standard.

After correcting resulting signal for the blank, graph the data showing such that a straight line results.

Show that the volume % of acetylene in the unknown can be found where the line crosses the ordinate.

Calculate the volume % of acetylene in the unknown.